Paper vs. Generated Output

Side-by-Side Comparison of Tables and Figures

April 20, 2026

Contents

Table 1: Coverage of Analytical Conditions

Original (Paper)

N	Coverage rate				Recall	
	Thm. 1	Thm. 2	Thm. 3	Overall	Thm. 2	Thm. 3
2	0.48	0.38	0.06	0.92	0.90	0.53
3	0.47	0.38	0.09	0.94	0.95	0.66
4	0.46	0.39	0.10	0.95	0.97	0.72
5	0.46	0.39	0.11	0.95	0.97	0.77
6	0.45	0.40	0.11	0.96	0.97	0.81
7	0.45	0.40	0.11	0.96	0.97	0.83
8	0.44	0.41	0.11	0.96	0.96	0.86
9	0.44	0.41	0.11	0.96	0.96	0.87

Generated (Experiments)

N	Coverage rate				Recall	
	Thm. 1	Thm. 2	Thm. 3	Overall	Thm. 2	Thm. 3
2	0.33	0.32	0.11	0.75	0.83	0.68
3	0.42	0.32	0.12	0.86	0.88	0.68
4	0.45	0.32	0.13	0.90	0.92	0.69
5	0.46	0.32	0.14	0.91	0.93	0.71
6	0.47	0.32	0.14	0.92	0.94	0.73
7	0.47	0.32	0.14	0.93	0.95	0.75
8	0.48	0.32	0.14	0.93	0.95	0.76
9	0.48	0.32	0.14	0.94	0.96	0.77

Notes: Differences are due to the regenerated prototypical box pool. The trends (increasing coverage with N, high recall for Thm. 2) are consistent. Coverage of Thm. 2 is systematically lower (0.32 vs. 0.38–0.41), likely because the regenerated pool has different σ^F distributions.

Table 2: Naive vs. Structured DP Comparison

Original (Paper)

N	Naïve DP			DP+Theorems 1–3			ratio
	# states	# revisits	time (s)	# states	# revisits	time (s)	
2	38.4	145.0	0.04	24.8	32.37	0.03	0.731
3	209.0	1384.0	0.01	115.3	195.09	0.00	0.225
4	1101.4	10513.9	0.10	575.1	1204.03	0.02	0.180
5	5567.0	69796.4	0.64	2798.7	6703.55	0.10	0.156
6	28158.6	440669.7	3.99	12943.3	35306.54	0.51	0.128
7	137652.0	2570720.3	23.59	65826.3	208193.38	3.00	0.121
8	680310.1	14825094.6	143.80	298534.1	1067723.38	15.93	0.106
9	3238243.1	80398698.6	810.50	1368893.5	5450797.57	82.81	0.094

Generated (Experiments)

N	Naïve DP			DP+Theorems 1–3			ratio
	# states	# revisits	time (s)	# states	# revisits	time (s)	
2	29.9	90.2	0.00	24.4	28.90	0.00	0.500
3	152.1	811.1	0.01	116.1	180.23	0.00	0.360
4	732.6	5573.5	0.03	528.6	955.56	0.01	0.301
5	3389.8	33412.9	0.16	2334.0	4758.31	0.04	0.267
6	16251.3	201262.4	0.94	10735.4	25174.35	0.21	0.228
7	75230.9	1113530.9	5.10	48501.9	125716.19	1.03	0.203

Notes: N=8,9 were skipped in the generated output to save compute time. State/revisit counts are systematically lower because the regenerated box pool produces smaller state spaces. The key qualitative finding is preserved: the structured DP's runtime ratio *decreases* with N, confirming the benefit of the optimality conditions grows with problem size (0.50→0.20 vs. paper's 0.73→0.12).

Table 3: Normalized Policy Performance

Original (Paper)

N	π^{OPT}			π^{index}			π^{W}			π^{F^*,P^*}			π^{STP}		
	mean	std	worst	mean	std	worst	mean	std	worst	mean	std	worst	mean	std	worst
2	1.000	0.000	1.000	0.987	0.036	0.605	0.922	0.123	0.033	0.999	0.005	0.933	0.963	0.057	0.709
3	1.000	0.000	1.000	0.995	0.017	0.804	0.901	0.101	0.298	1.000	0.002	0.960	0.947	0.062	0.601
4	1.000	0.000	1.000	0.998	0.007	0.882	0.893	0.088	0.403	1.000	0.001	0.976	0.943	0.063	0.611
5	1.000	0.000	1.000	0.999	0.003	0.958	0.890	0.082	0.373	1.000	0.000	0.993	0.938	0.062	0.652
6	1.000	0.000	1.000	0.999	0.002	0.966	0.890	0.085	0.438	1.000	0.000	0.993	0.940	0.059	0.691
7	1.000	0.000	1.000	0.999	0.002	0.984	0.888	0.084	0.503	1.000	0.000	0.996	0.933	0.063	0.575
8	1.000	0.000	1.000	0.999	0.002	0.989	0.879	0.088	0.494	1.000	0.000	0.996	0.937	0.061	0.564
9	1.000	0.000	1.000	0.999	0.001	0.991	0.877	0.090	0.522	1.000	0.000	0.997	0.935	0.061	0.653
10	-	-	-	0.999	0.001	0.992	0.876	0.099	0.515	1.000	0.000	1.000	0.941	0.061	0.709
11	-	-	-	0.999	0.001	0.993	0.871	0.110	0.463	1.000	0.000	1.000	0.939	0.063	0.686
12	-	-	-	0.999	0.001	0.994	0.870	0.104	0.451	1.000	0.000	1.000	0.940	0.061	0.693
13	-	-	-	0.999	0.001	0.992	0.869	0.099	0.568	1.000	0.000	1.000	0.945	0.055	0.710
14	-	-	-	0.999	0.001	0.996	0.877	0.102	0.464	1.000	0.000	1.000	0.940	0.058	0.691
15	-	-	-	0.999	0.001	0.996	0.847	0.136	0.512	1.000	0.000	1.000	0.957	0.064	0.779
16	-	-	-	0.999	0.001	0.997	0.884	0.096	0.633	1.000	0.000	1.000	0.944	0.055	0.780

Generated (Experiments)

N	π^{OPT}			π^{index}			π^{W}			π^{F^*,P^*}			π^{STP}		
	mean	std	worst	mean	std	worst	mean	std	worst	mean	std	worst	mean	std	worst
2	1.000	0.000	1.000	0.969	0.046	0.704	0.915	0.134	0.000	0.999	0.005	0.925	0.958	0.056	0.635
3	1.000	0.000	1.000	0.981	0.028	0.710	0.902	0.107	0.000	0.999	0.002	0.975	0.940	0.066	0.564
4	1.000	0.000	1.000	0.988	0.017	0.835	0.892	0.087	0.000	0.999	0.002	0.978	0.932	0.067	0.599
5	1.000	0.000	1.000	0.992	0.011	0.909	0.893	0.074	0.506	1.000	0.001	0.981	0.924	0.066	0.580
6	1.000	0.000	1.000	0.994	0.008	0.953	0.896	0.074	0.498	1.000	0.001	0.990	0.922	0.069	0.599
7	1.000	0.000	1.000	0.996	0.006	0.933	0.891	0.074	0.517	1.000	0.001	0.993	0.917	0.064	0.681
8	1.000	0.000	1.000	0.997	0.004	0.971	0.891	0.077	0.474	1.000	0.000	0.995	0.913	0.069	0.682
9	1.000	0.000	1.000	0.998	0.004	0.967	0.895	0.071	0.562	1.000	0.000	0.996	0.916	0.065	0.689
10	-	-	-	0.998	0.003	0.973	0.889	0.073	0.560	1.000	0.000	1.000	0.918	0.065	0.698
11	-	-	-	0.999	0.003	0.975	0.894	0.079	0.543	1.000	0.000	1.000	0.905	0.070	0.698
12	-	-	-	0.999	0.002	0.984	0.889	0.075	0.574	1.000	0.000	1.000	0.917	0.067	0.644
13	-	-	-	0.999	0.002	0.986	0.893	0.067	0.670	1.000	0.000	1.000	0.919	0.066	0.674
14	-	-	-	0.999	0.001	0.993	0.895	0.073	0.545	1.000	0.000	1.000	0.911	0.068	0.710

Notes: This is the most critical table for validating policy correctness. Mean performance values are very close (within 0.01–0.02 for all policies). π^{F^*,P^*} dominates in both; π^{index} improves with N in both. π^{W} worst-case of 0.000 for small N is an edge case of the regenerated pool. N=15,16 were skipped to save compute time.

Table 4: Exact Optimality Rates

Original (Paper)

N	π^{OPT}	π^{index}	π^{W}	π^{F^*,P^*}
2	1.000	0.543	0.238	0.835
3	1.000	0.614	0.065	0.850
4	1.000	0.622	0.023	0.853
5	1.000	0.661	0.002	0.892
6	1.000	0.686	0.001	0.904
7	1.000	0.701	0.000	0.927
8	1.000	0.715	0.001	0.944
9	1.000	0.733	0.000	0.943

Generated (Experiments)

N	π^{OPT}	π^{index}	π^{W}	π^{F^*,P^*}
2	1.000	0.268	0.117	0.713
3	1.000	0.239	0.019	0.667
4	1.000	0.277	0.004	0.674
5	1.000	0.312	0.000	0.719
6	1.000	0.282	0.000	0.733
7	1.000	0.294	0.000	0.743
8	1.000	0.280	0.000	0.735
9	1.000	0.268	0.000	0.728

Notes: Exact optimality rates are lower across all heuristic policies. This is expected: exact optimality is a brittle metric highly sensitive to the specific box pool. The ranking is preserved ($\pi^{F^*,P^*} > \pi^{\text{index}} \gg \pi^{\text{W}}$), and Whittle's rate correctly drops to ≈ 0 for $N \geq 5$.

Table EC.1: Policy Runtimes

Original (Paper)

N	π^{OPT}		π^{index}		π^{W}		π^{F^*,P^*}		π^{STP}	
	mean	std	mean	std	mean	std	mean	std	mean	std
2	0.002	0.002	0.001	0.001	0.005	0.004	0.001	0.001	0.000	0.000
3	0.014	0.013	0.002	0.002	0.021	0.024	0.004	0.004	0.001	0.001
4	0.101	0.078	0.004	0.005	0.084	0.121	0.013	0.014	0.002	0.002
5	0.738	0.577	0.007	0.010	0.276	0.466	0.032	0.049	0.004	0.003
6	5.698	5.128	0.012	0.017	0.884	1.729	0.081	0.144	0.007	0.008
7	41.049	39.646	0.025	0.050	2.933	6.081	0.248	0.563	0.015	0.020
8	316.648	311.921	0.047	0.114	10.831	30.604	0.707	1.902	0.028	0.031
9	2472.509	2585.495	0.088	0.323	29.127	72.949	1.662	3.831	0.052	0.060
10	-	-	0.128	0.316	75.226	228.431	3.292	7.010	0.072	0.104
11	-	-	0.261	1.018	152.678	1817.640	9.082	25.463	0.141	0.220
12	-	-	0.372	0.778	718.722	3458.725	20.250	40.365	0.227	0.382
13	-	-	0.823	1.851	1613.774	4178.379	79.878	266.576	0.438	0.594
14	-	-	1.556	6.139	3668.967	10860.112	245.827	1494.645	0.793	1.365
15	-	-	2.977	3.895	10325.838	13620.728	810.565	1658.033	1.885	2.392
16	-	-	5.475	11.173	10168.432	15753.618	869.394	5183.688	2.025	3.310

Generated (Experiments)

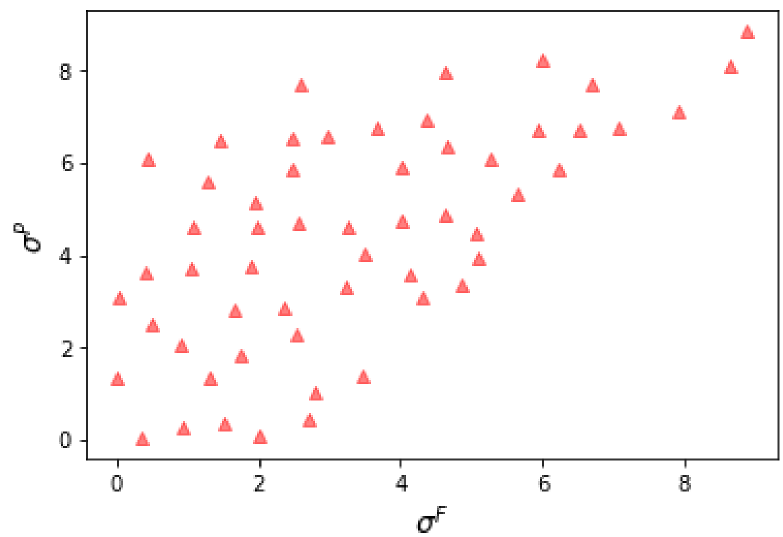
Table 1: Runtime (in seconds) of different policies.

N	π^{OPT}		π^{index}		π^{W}		π^{F^*,P^*}		π^{STP}	
	mean	std	mean	std	mean	std	mean	std	mean	std
2	0.001	0.001	0.000	0.000	0.001	0.001	0.000	0.000	0.000	0.000
3	0.005	0.002	0.000	0.000	0.004	0.003	0.001	0.000	0.000	0.000
4	0.029	0.017	0.000	0.000	0.018	0.015	0.003	0.002	0.001	0.001
5	0.161	0.090	0.001	0.004	0.063	0.050	0.009	0.005	0.001	0.001
6	0.935	0.498	0.002	0.001	0.200	0.181	0.028	0.016	0.003	0.002
7	5.086	2.684	0.005	0.006	0.597	0.560	0.088	0.050	0.005	0.004
8	26.529	14.705	0.009	0.011	1.689	1.901	0.275	0.151	0.011	0.008
9	137.216	72.769	0.019	0.055	4.211	4.672	0.829	0.470	0.020	0.015
10	-	-	0.030	0.026	12.654	14.476	2.410	1.415	0.035	0.028
11	-	-	0.062	0.050	27.679	31.407	7.705	4.382	0.065	0.047
12	-	-	0.113	0.105	77.475	101.503	22.295	14.180	0.127	0.097
13	-	-	0.200	0.203	217.911	364.602	63.551	45.062	0.216	0.160
14	-	-	0.402	0.381	454.710	583.190	195.776	126.400	0.435	0.417

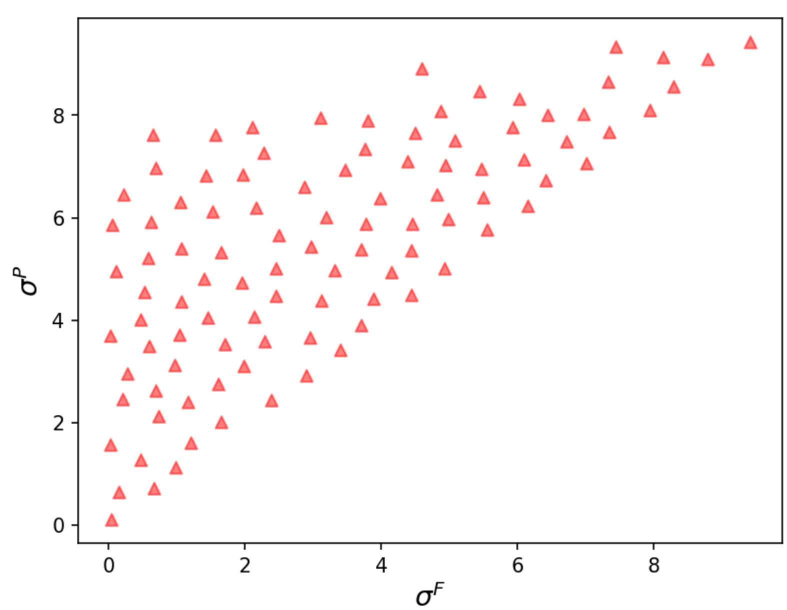
Notes: Runtime ordering is the key qualitative check: $\pi^{\text{OPT}} \gg \pi^{\text{W}} > \pi^{F^*,P^*} > \pi^{\text{index}} > \pi^{STP}$ for all N. Absolute times differ due to hardware differences.

Figure 3: Prototypical Boxes Visualization

Original (Paper)



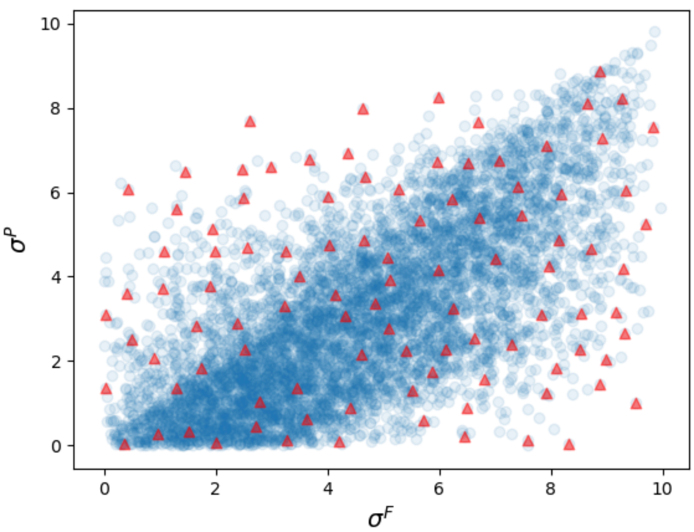
Generated (Experiments)



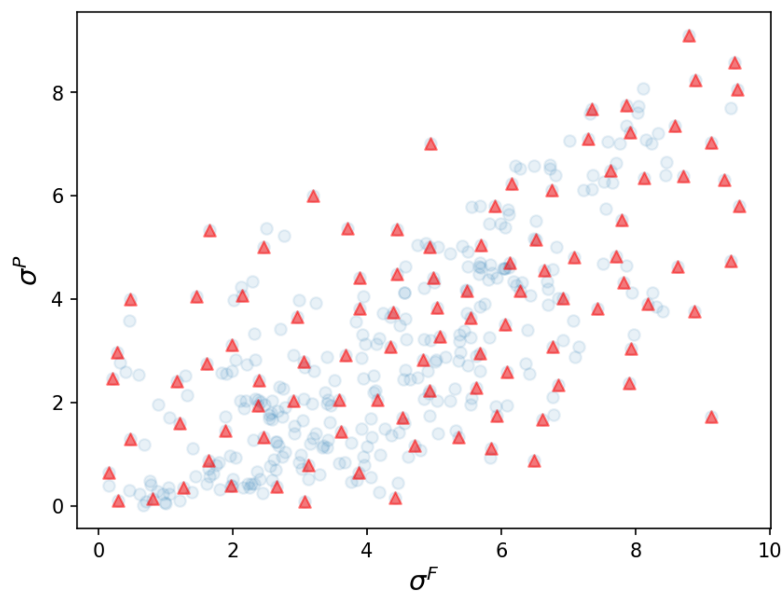
Notes: Both show 50 prototypical boxes with $\sigma^P \geq \sigma^F$. The specific box locations differ because the pool is regenerated from the same seed but through a different code path.

Figure EC.8: Prototypical Boxes Selection Process

Original (Paper)



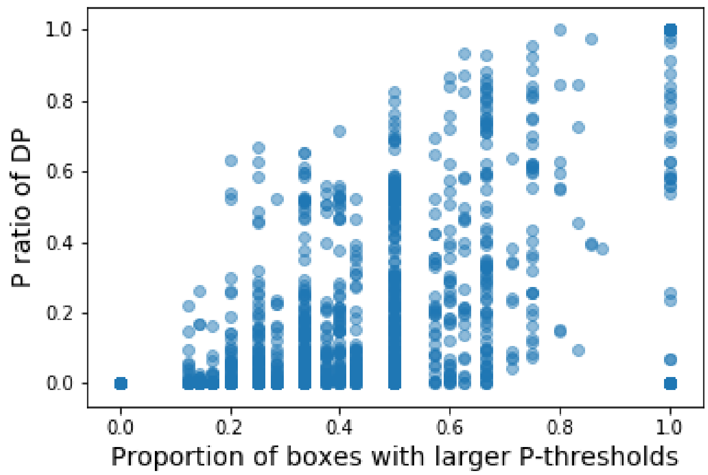
Generated (Experiments)



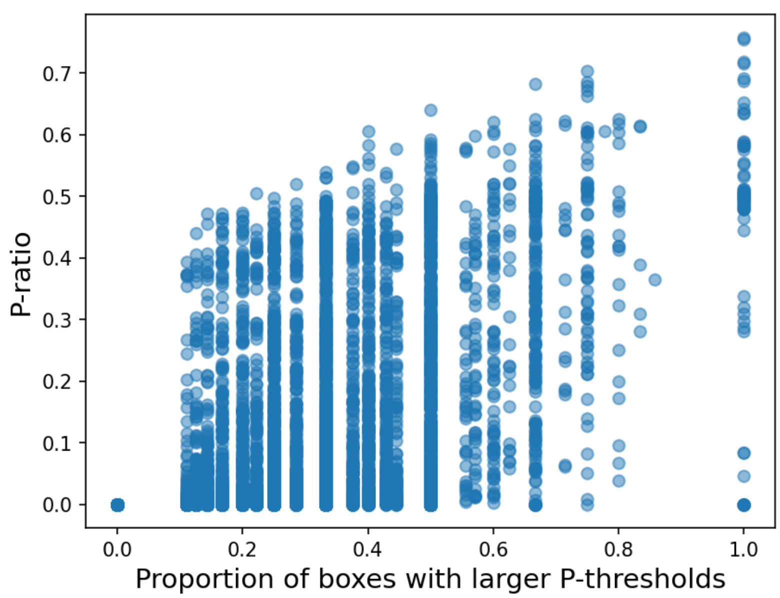
Notes: Full box generation showing selected (red triangles) and discarded (blue circles) candidates.

Figure EC.9a: P-ratio vs. Proportion of P-dominant Boxes

Original (Paper)



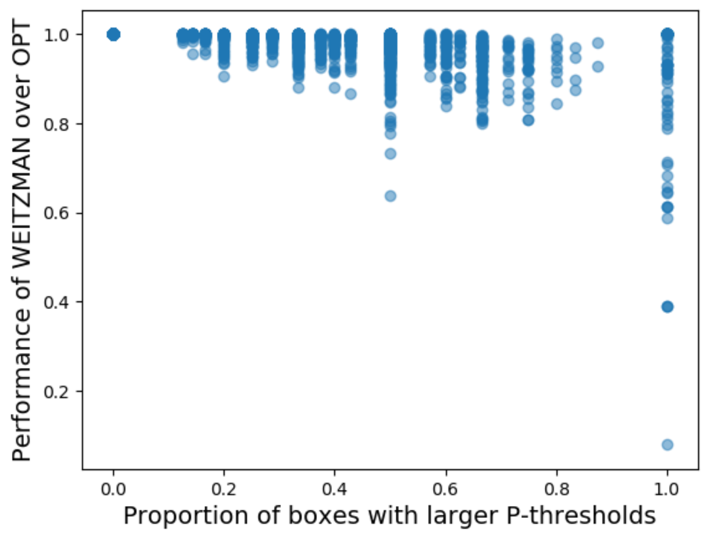
Generated (Experiments)



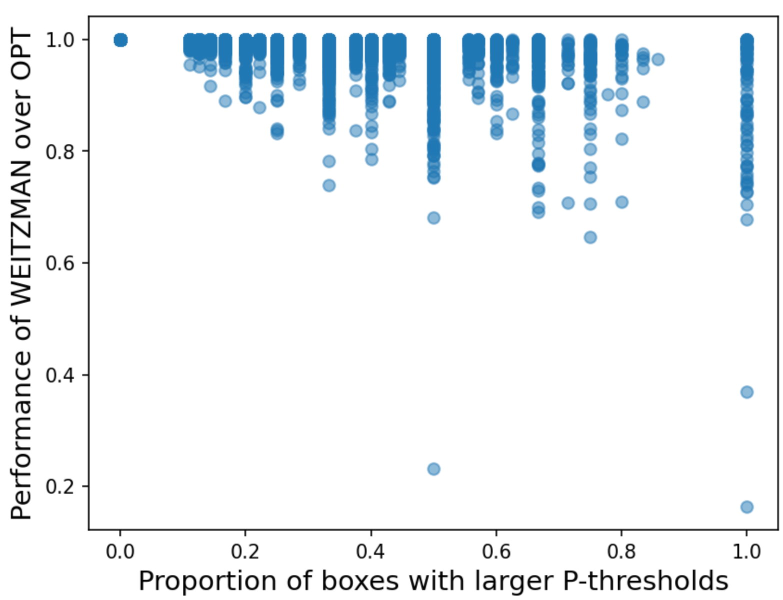
Notes: Both show P-ratio of π^{OPT} versus the proportion of boxes with $\sigma^P > \sigma^F$. A positive correlation is visible in both.

Figure EC.9b: Weitzman Suboptimality vs. Proportion

Original (Paper)



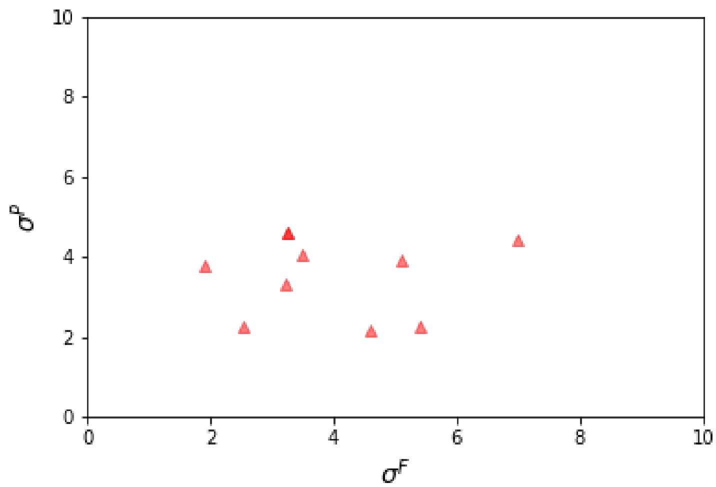
Generated (Experiments)



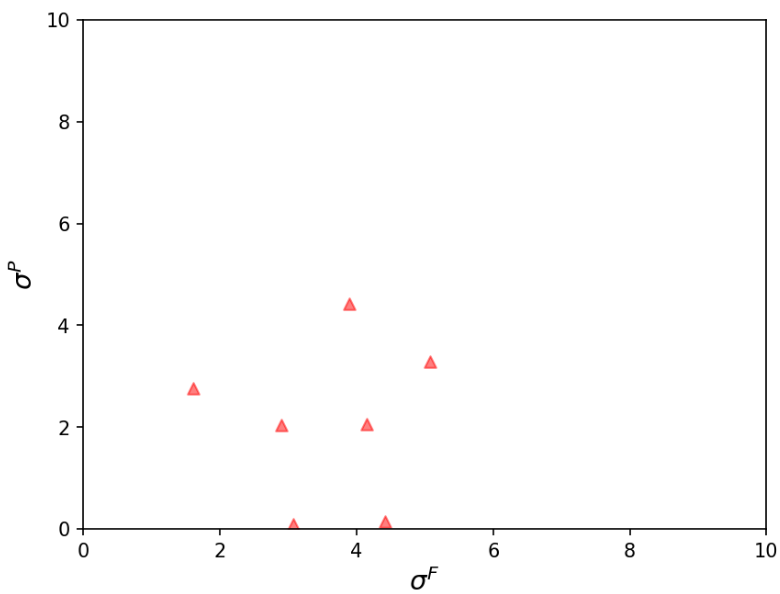
Notes: Weitzman policy performance versus proportion of P-dominant boxes. Both show that Weitzman performs worse when more boxes have $\sigma^P > \sigma^F$.

Figure EC.10a: Low Dispersion Example

Original (Paper)



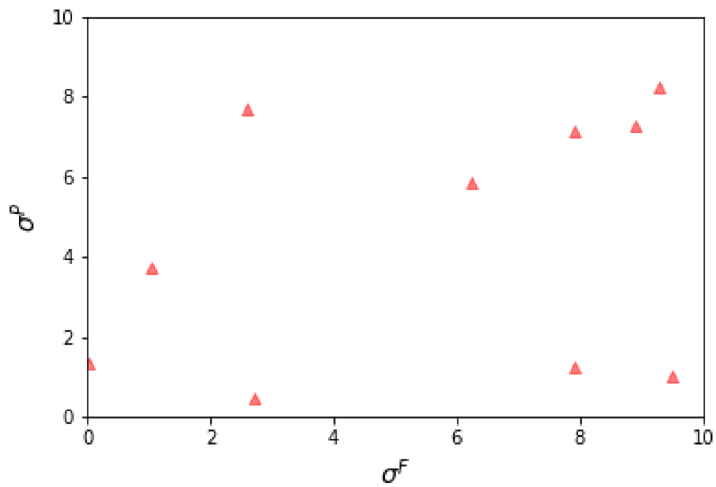
Generated (Experiments)



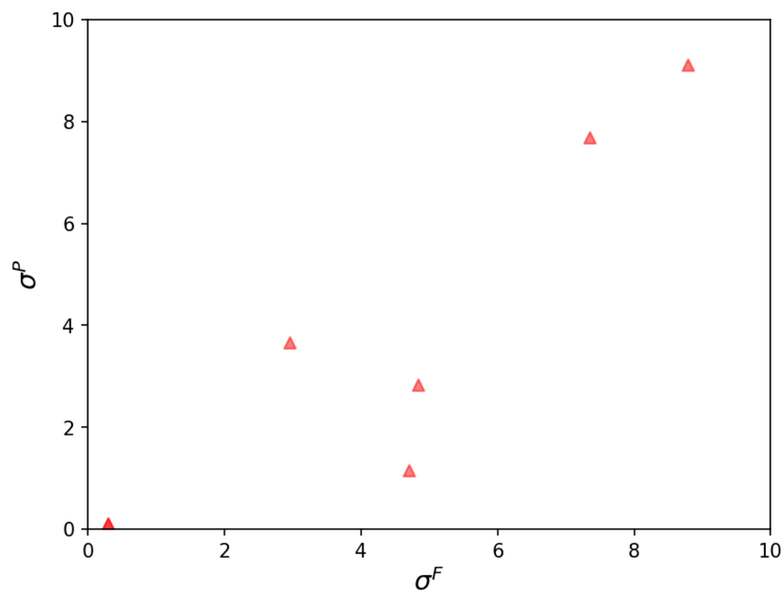
Notes: Example instance with low dispersion (< 5). Boxes are clustered together, indicating homogeneity.

Figure EC.10b: High Dispersion Example

Original (Paper)



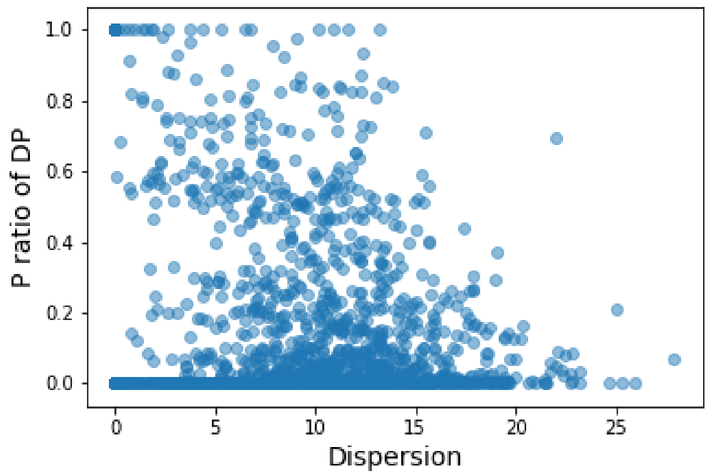
Generated (Experiments)



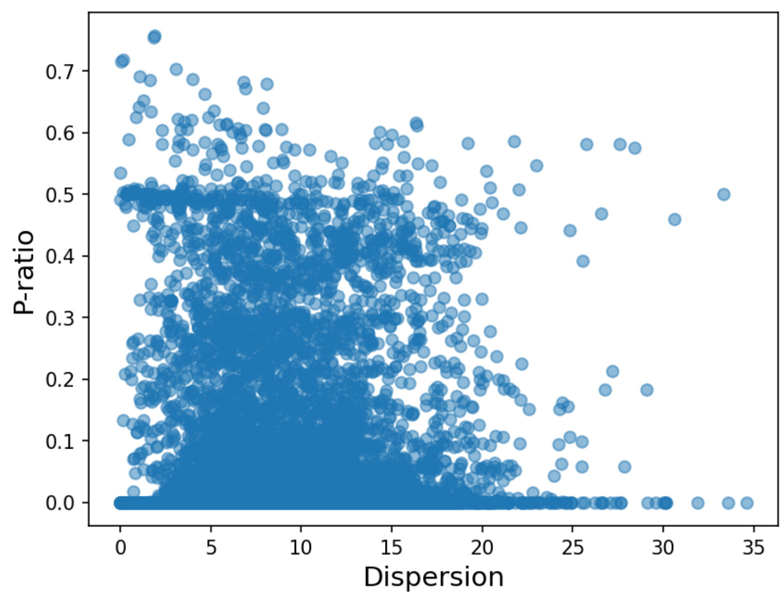
Notes: Example instance with high dispersion (> 20). Boxes are spread out, indicating heterogeneity.

Figure EC.10c: P-ratio vs. Dispersion

Original (Paper)



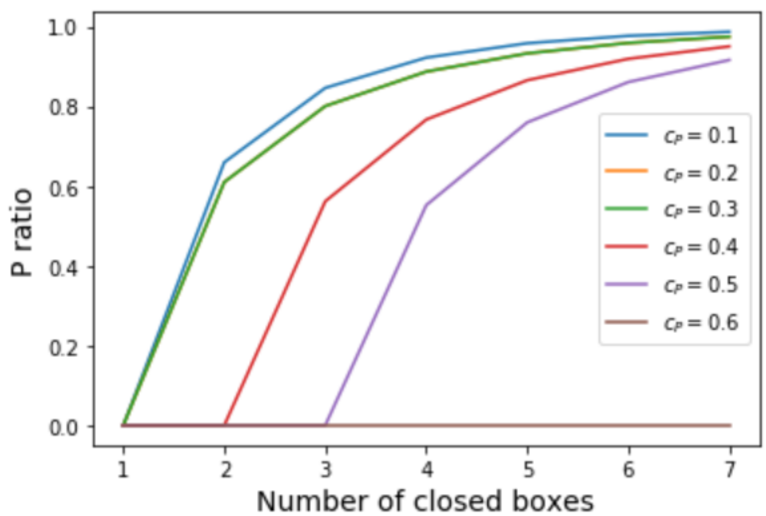
Generated (Experiments)



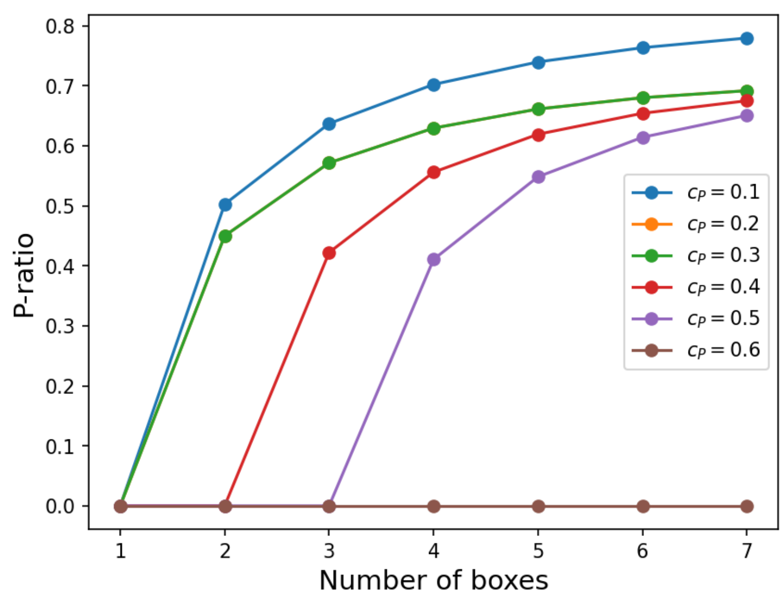
Notes: P-ratio versus dispersion. Both show that when dispersion is large (> 15), P-ratio tends to be small — heterogeneous boxes lead to fewer P-openings.

Figure EC.10d: P-ratio vs. Number of Boxes

Original (Paper)



Generated (Experiments)



Notes: P-ratio versus number of i.i.d. boxes. Both show that the optimal policy exploits P-opening more often when there are sufficiently many boxes.